Abstract

Predicting species range dynamics under rapid global change remains a central challenge in conservation biology, invasive species management, and biodiversity forecasting. Dispersal is a key driver of range shifts, yet existing modeling approaches face a trade-off between realism and scalability: individual-based models can explicitly account for landscape heterogeneity, mortality and genetic effects but are computationally intensive and data-hungry, whereas kernel-based colonization–extinction models scale well but typically rely on isotropic dispersal kernels that ignore landscape structure. Here, we introduce a low-parametrization, spatially explicit colonization–extinction model that bridges this gap by embedding landscape connectivity, mortality, and growth dynamics in a single framework using anisotropic random walk kernels. The dispersal component requires only a single species-level input—the mean natal dispersal distance—which can be estimated from allometric relationships, while mortality, path selection, and growth rates are inferred directly from spatial occurrence patterns and habitat suitability maps. By integrating dispersal and growth within a relative-abundance framework, the model explicitly accounts for sampling bias in opportunistic occurrence data. We demonstrate the approach on three contrasting case studies—the recent expansion of the Black-winged Kite (*Elanus caeruleus*), the assisted recolonization of the European Beaver (*Castor fiber*) in France and the passive recolonization of the Eurasian wolf (*Canis lupus*) across Italy and Alps—showing that the model reliably reconstructs observed expansion speeds, spatial constraints, and major landscape barriers at continental scales. Our results highlight the potential of this parsimonious, connectivity-aware framework to support large-scale biodiversity projections and conservation planning in data-limited contexts.

February 19, 2026

PARADIS: A minimal parametrization model to predict future species range via growth-dispersal joint estimation from occurrences pattern.

1 Introduction 1

Modeling species range shifts is a central challenge in conservation biology (Sarkar et al. 2006), invasive species management (Colberg et al. 2024) , and extinction risk assessment (Weiss-Lehman and Shaw 2020). Under ongoing human-induced global change (Pörtner et al. 2022), species distributions are increasingly reshaped at unprecedented rates. These distributional changes can destabilize ecological communities, triggering trophic cascades (Borrvall and Ebenman 2006), local or global extinctions (Petchey et al. 2008), and reductions in ecosystem services (Staddon et al. 2010). Anticipating future species ranges is therefore essential for designing effective conservation strategies and for forecasting future ecosystem functioning (Kass et al. 2024).

Dispersal is a key process underlying species range dynamics and constitutes a major focus of theoretical and applied modeling efforts. Dispersal refers to the movement of organisms or their propagules through space (Clobert et al. 2012), including seeds in plants and individuals themselves in animals. Dispersal models generally describe the redistribution of propagules from their natal location to their eventual establishment, either as a fixed location for plants or as a home range for animals (Penteriani and Delgado 2009).

Species range dynamics can be modeled at multiple spatial and temporal scales (Yang et al. 2021; Engler, Hordijk, and Guisan 2012; Robledo-Arnuncio et al. 2014). Among existing approaches, individual-based models (IBMs), derived from a Lagrangian framework (Baguette, Stevens, and Clobert n.d.), explicitly represent all phases of dispersal, including emigration, transience, and settlement (Bowler and Benton 2005). These models simulate the

Table 1: Notations

Symbol	Meaning	Range	Typical values	Nature
MDD	mean dispersal distance	$\mathbb{R}^+$	3 - 50 km/gen	given
h	mean habitat suitability	$[0, 1]$	0.7 - 0.90	estimated
E_w	expected number of steps in the random walk	$\mathbb{R}^+$	100 - 3000 steps	learned
E_w^*	expected number of step accounting mortality	$\mathbb{R}^+$	-	estimated
n	displacement selectivity	$\mathbb{R}^+$	1000-3000	learned
r	risk scaling	$[0, r_{max}]$	0.001 - 0.00001	learned
g	growth rate	$\mathbb{R}^+$	0.001 - 0.1	learned
i	location on the grid	-	-	-
j	location on the grid	-	-	-
$p(i \to j)$	transition rate between location i and j	[0,1]	-	learned
n_l	total number of locations	$\mathbb{N}$	-	-
$\mathbf{K}$	matrix where rows contain location kernels for each starting location	$n_l \times n_l$	-	learned
$\mathbf{W}$	adjacency matrix containing transition rates between locations	$n_l \times n_l$	-	learned
$[i, j]$	entry in a matrix where is the transition rate $i \to$j	-	-	-
$\Delta N_i(t)$	offspring production at time t and position i	$\mathbb{R}^+$	-	unknown
$N_i(t)$	abundance at time t and position i	$\mathbb{R}^+$	-	unknown
C_i	carrying capacity at position i	$\mathbb{R}^+$	-	unknown
C_{taxa}	total abundance	$\mathbb{R}^+$	-	unknown
$\tilde{N}_i(t)$	relative abundance at time t and position i	[0,1]	0. - 0.5	simulated
$\tilde{C}_i$	relative carrying capacity at position i	[0,1]	0.001 - 0.5	estimated
$n_{sp,i}$	number of species occurrences at location i	$\mathbb{N}$	0 - 200 obs	given
$n_{taxa,i}$	number of taxa occurrences at location i	$\mathbb{N}$	0 - 1000 obs	given
$\tilde{n}_i$	observed relative abundance at location i	[0,1]	0. - 0.5	estimated

movement of individual organisms through small spatial steps, often using correlated random walks (Johnson et al. 2008), and incorporate behavioral components such as perceptual range, directional persistence, and local movement decisions (Coulon et al. 2015). Several user-friendly implementations exist, including the R package RangeShiftR (Malchow, Bocedi, et al. 2021), which provides dispersal modules such as the Stochastic Movement Simulator (SMS) [Palmer et al. 2011].

A major strength of individual-based approaches lies in their ability to explicitly incorporate landscape heterogeneity. Movement is constrained by spatial variation in terrain characteristics, where some landscape elements facilitate movement while others impose resistance or risk. This heterogeneity is commonly represented using resistance or cost surfaces (Etherington 2016). For example, road networks can act as strong barriers to movement, reducing dispersal fluxes between habitat patches (Shepard et al. 2008). Because IBMs explicitly track individual states (Baguette, Stevens, and Clobert n.d.), they can naturally integrate such local resistance effects (Chevy et al. 2025). However, these models require extensive parameterization and incur high computational costs in terms of runtime and memory. As a result, they are most suitable for small-scale studies on well-documented species, and their applicability is limited for continental-scale simulations or for poorly known taxa such as invertebrates, amphibians, reptiles, and small mammals (Titley, Snaddon, and Turner 2017; Verde Arregoitia 2016).

An alternative perspective is provided by Eulerian approaches (Baguette, Stevens, and Clobert n.d.), notably kernel-based dispersal models. Rather than simulating individual trajectories, these methods describe dispersal statistically using probability density functions known as dispersal kernels (Clobert et al. 2012). Kernels represent the redistribution of population density across space during a given time step, simultaneously integrating all phases of dispersal for multiple individuals. When explicitly defined in space and time, these kernels are referred to as location kernels. Coupled with population dynamic models, they form the basis of colonization–extinction frameworks that simulate spatio-temporal changes in species distributions. Several R packages implement such approaches, including RangeShiftR (Malchow, Bocedi, et al. 2021), MigClim (Engler, Hordijk, and Guisan 2012), CATS (Gattringer et al. 2023), megaSDM (Shipley et al. 2022), and more recently implementations in Julia (Blechschmidt and Cabral 2025).

In most kernel-based models, dispersal kernels are defined a priori using species-specific information such as mean dispersal distance, and are applied uniformly across the landscape at each time step. Such isotropic kernels have proven effective for modeling ballistic seed dispersal in plants (Bullock et al. 2017). However, isotropic assumptions are often unrealistic in heterogeneous landscapes, as they ignore spatial variation in movement constraints. This limitation has motivated the development of anisotropic kernels, which incorporate directional biases induced by wind, river flow, or other landscape features (Rieux et al. 2014; Acevedo et al. 2015; Savage et al. 2011; Putten et al. 2012). Anisotropy is also required to represent complex movement trajectories arising from habitat selection during the transient phase in animals (De Boer et al. 2023), which can indirectly influence plant dispersal through zoochory (D'hondt et al. 2012). Incorporating landscape resistance into kernel-based methods remains challenging (Unnithan Kumar et al. 2022), as these models do not represent individual paths but instead describe the statistical outcome of dispersal events.

Least Cost Path (LCP) analysis, initially developed in the context of circuit theory to assess landscape connectivity (Dickson et al. 2019), has become a widely used method to generate anisotropic dispersal kernels (Compton et al. 2007). In landscape ecology, connectivity refers to the presence of corridors—paths that are practicable for dispersers. LCP algorithms typically rely on Dijkstra's breadth-first search algorithm, although alternative formulations such as conditional minimum transit costs or multiple shortest paths have been proposed [Pinto and Keitt 2009]. These algorithms incrementally accumulate resistance values to identify paths that minimize movement cost from a source location (Etherington and Penelope Holland 2013). The resulting cumulative resistance map assigns to each pixel the cost of the least costly path from the origin. LCP-based approaches implicitly assume that dispersers follow least-effort paths (Driezen et al. 2007) to optimize energy expenditure during movement (De Boer et al. 2023), and that they possess full knowledge of landscape structure prior to dispersal (Balbi et al. 2019). While these assumptions may hold in some contexts, they have been questioned in others, especially during natal dispersal (Eycott et al. 2012).

Connectivity analyses derived from LCPs are inherently static and do not explicitly incorporate temporal information. They identify accessible corridors but do not estimate the time required for dispersers to traverse them, limiting their direct applicability in colonization–extinction models. Nonetheless, spatio-temporal anisotropic kernels, termed least cost

kernels, can be derived by introducing temporal calibration. This requires estimating the cumulative resistance that individuals can traverse within a given time step, typically using telemetry or capture–mark–recapture data (Etherington, Perry, et al. 2014). The resulting dispersal capacity is then translated into a location kernel by fitting a distribution over the cumulative resistance map. In data-poor situations, a simplified calibration may be performed by matching known mean dispersal distances to corresponding cumulative resistance values, provided a consistent relationship exists between Euclidean distance and least-cost distance in the landscape (Foltête et al. 2015).

Despite their widespread use, LCP-based kernels suffer from several limitations. They rely on a single optimal path to estimate dispersal fluxes, potentially underestimating contributions from multiple suboptimal routes. This single-path assumption also makes LCP outputs highly sensitive to spatial resolution, neighborhood definition, and local variations in resistance values, particularly in bottlenecked landscapes (Etherington 2016). Moreover, LCP analyses are strongly dependent on the accuracy of resistance layers. While recent approaches infer resistance implicitly from habitat suitability models, which outperform earlier expert-based methods [Milanesi 2017], uncertainty remains. Mortality is also not explicitly incorporated into LCP-based dispersal, except indirectly through calibration data (Etherington 2016), and must therefore be accounted for separately in population dynamic models.

More recently, Spatial Absorbing Markov Chain (SAMC) models have been proposed as an alternative framework for dispersal modeling (Fletcher et al. 2023). Like LCP approaches, SAMC relies on resistance surfaces, but it represents the landscape as a graph in which nodes correspond to spatial locations and dispersal states, and edges represent probabilistic transitions. Unlike LCP-based methods, SAMC considers all possible movement paths simultaneously rather than focusing on a single least-cost route. This multi-path formulation reduces sensitivity to spatial resolution and neighborhood structure, accounts for concentration effects, and allows mortality to be explicitly incorporated via a risk layer. By relaxing the assumption of perfect landscape knowledge prior to dispersal, SAMC is particularly well suited to modeling natal dispersal. In addition to connectivity assessment, SAMC can generate spatio-temporal location kernels, referred to as random walk kernels.

Despite these advances, the practical application of SAMC remains limited by its complex calibration requirements, particularly for large-scale conservation planning and continental-

scale ecological analyses. Such applications often require colonization–extinction models with simplified dispersal components that retain sensitivity to landscape connectivity while minimizing parameterization demands. In parallel, the demographic component of these models presents additional challenges: functional relationships between species growth rates and environmental conditions are poorly known in most cases (Hirzel and Le Lay 2008), complicating the mechanistic simulation of abundance-based range dynamics. To circumvent this issue, some models propagate directly "presence", as in MigClim, or use dispersal-based spatial filters to constrain occupancy models [Shipley et al. 2022]. However, these simplifications hinder the representation of nonlinear dynamics and temporal aspects of colonization, and dispersal buffers derived from individual movement rates do not necessarily reflect population-level expansion processes.

To address these limitations, we propose a spatially explicit, easy-to-calibrate colonization–extinction model designed for data-scarce contexts and large-scale applications. The dispersal component of this model can be interpreted as a specific case of SAMC and requires only a single species-specific parameter: the mean natal dispersal distance. However, this quantity can be estimated from allometric relationships (Sutherland et al. 2000; Weeks et al. 2022; Santini et al. 2013). Mortality risk, dispersal behavior, and growth rates are all jointly inferred directly from spatial occurrence patterns and habitat suitability maps, drastically reducing number of parameters. Such low-parameter approaches are essential for predicting large-scale biodiversity patterns under global change, both for conservation management and for advancing theoretical ecology. Here, we present the structure, results, limitations, and potential applications of this colonization–extinction modeling framework.

2 Methods

2.1 Dispersal model derivation

We consider the study landscape as a discrete grid, where individuals can disperse following transition rates encoded in an adjacency matrix. To derive the local transition rates between two locations, we consider survival, dispersal capacities and path selection. During a step from one location i to another adjacent one j -we choose Moore neighborhood definition- we express the transition rate in the following form:

$$p(i \to j) = \mathbb{P}(Continue\ dispersal)\mathbb{P}(Choose\ i \to j)\mathbb{P}(Survive\ i \to j)$$

For model tractability, we choose to model the probability to continue dispersal as a geometric law, where the expected number of step for the walk before establishment (E_w) permits to describe the dispersal capability of the species. The rate for such behavior should be of the form:

$$\mathbb{P}(Continue\ dispersal) = \frac{E_w}{1 + E_w}$$

It can be viewed as 1 - $\mathbb{P}(Establishment)$.

The probability to survive the chosen transition $\mathbb{P}(Survive\ i \to j)$ need to incorporate the specific location conditions, either favorable or deleterious for the species. If the species crosses an unfavorable environment we expect the mortality to increase. Habitat suitability models permits to produce a statistic in [0,1] summarizing the similarity of an environmental condition to the ones where the species is commonly observed. The fundamental assumption in connectivity analysis is to assume that types of environments used the most by a species are also environments where they exhibit the smaller mortality and displacement resistance (Keeley, Beier, and Gagnon 2016). To map the habitat suitability statistic h to a probability and allow for a large range of possibilities while keeping it simple, we expressed the survival probability as a power-law of habitat suitability:

$$\mathbb{P}(Survive\ i \to j) = h_{i,j}^r$$

The risk scaling factor r scales the value of h and maps it to the unknown survival rate, a high r corresponds to a low survival rate overall except for high values of h, and a small r means that the survival rate increase sharply with a better habitat suitability h. The r parameter role can be compared to the translation step of habitat quality into cost surface in classical dispersal connectivity framework (Keeley, Beier, and Gagnon 2016)(Etherington 2016).

Due to the same fundamental assumption, species, especially in animals, are expected to be able to select paths in their local environments to reduce the risk of dying during transience. Such decision should be made by comparing local transitions associated risks (h^r) :

$$\mathbb{P}(Choose\ i \to j) = \frac{h_{i,j}^{nr}}{\sum\limits_{k \in M} h_{i,k}^{nr}}$$

n scale this risk and map it to a relative weight. Higher n gives more relative weight to the paths with better h compared to others in the neighborhood M, and smaller n gives more homogeneous path selection. We can see this as:

$$lim_{n \to 0} \frac{h_{i,j}^{nr}}{\sum\limits_{k \in M} h_{i,k}^{nr}} = \frac{1}{|M|}$$

Where $|M|$ denotes cardinality of the chosen neighborhood, in our case 4 -up, down, right, left-. In the case $n = 0$ the transitions are typically those of a random walk with mortality. The n parameter can be viewed as the reluctance to cross risky habitats. In risky landscapes it decreases propensity to leave favorable habitats, a well-known behaviors induced by habitat fragmentation on dispersal (see. (Schtickzelle, Mennechez, and Baguette 2006) for an example on Fritillary butterfly), but in highly structured habitats, it increases movement persistence.

We end with equation (0) as our expression of the transition rate between two locations i and j:

$$p(i \to j) = \frac{E_w}{1 + E_w} \cdot \frac{h_{i,j}^{nr}}{\sum\limits_{k \in M} h_{i,k}^{nr}} \cdot h_{i,j}^{r} \qquad (0)$$

All these transitions rates are encoded in an adjacency matrix $\mathbf{W}$ of size $n_l \times n_l$ where n_l is the number of locations on the map . Such that:

$$\mathbf{W}[i, j] = p(i \to j)$$

This structure permits to retrieve all dispersal kernels for each starting location by integrating over all possible path length the adjacency matrix $\mathbf{W}$, for a length k , k transitions has occurred ($\mathbf{W}^k$) and the propagule establishes one time ($1 - \frac{E_w}{1+E_w}$) :

$$\mathbf{K} = \sum_{k=0}^{+\infty} \left(1 - \frac{E_w}{1 + E_w}\right) \mathbf{W}^k$$

Knowing that the geometric series converges here cause the spectral radius of $\mathbf{W}$ is less than one:

$$\mathbf{K}(i \to j) = \frac{(\mathbf{I} - \mathbf{W})^{-1}[i, j]}{1 + E_w} \qquad (1)$$

Where $[i, j]$ denotes the ith row and j^{th} column of the considered matrix. Note that rows of matrix $\mathbf{K}$ do not sum to one, since mortality is directly blended inside the definition of the transition rate (see. Eq.0). $\mathbf{K}[i, j]$encodes the fraction of mass that started in i and arrived in j after dispersion, considering landscape structure, dispersal capacities and mortality along all possible paths. Row i of $\mathbf{K}$ therefore encodes the entire distribution of ending locations from starting point i. We can deduce new abundance distribution after dispersal by applying the matrix $\mathbf{K}$ on the right of an abundance row vector $\vec{N}$ with size $1 \times n_l$.

2.2 Growth model derivation

Alongside dispersal of newly produced individuals, density of occupied locations changes through time due to natural processes, birth and death. We consider a simple logistic model of population growth from a modified Verhulst's model. Where the density of the species at location i changes following:

$$\Delta N_{i,t} = g N_{i,t} \left(\frac{C_i - N_{i,t}}{C_i} \right)$$

Where $\Delta N_{i,t}$ is the change of abundance at step t and site i , $N_{i,t}$, the number of individuals occupying site i at time t , C_i the carrying capacity of site i, and finally g the growth rate of the species. When the population growth is negative $\Delta N_{i,t} < 0$, we consider that no new dispersers are produced, and the number of individuals occupying location i diminishes, however when $\Delta N_{i,t} > 0$, new individuals are subjected to a natal dispersion following the dispersal rules derived in the previous section. For an entire vector of abundances $\vec{N}_t$ of size $1 \times n_l$ where abundance at site i is encoded in the ith column:

$$G_t = g\vec{N}_t \odot (\vec{C} - \vec{N}_t) \oslash \vec{C}$$

Where $\odot$ is the pairwise vector multiplication (Hadamard product), and $\oslash$ the pairwise vector division. We denote by $G_t^+ = max(G_t, 0)$ the element-wise positive part of G_t. We similarly denote G_t^- as its negative part: $G_t^- = max(-G_t, 0)$. In this form, one step of the colonization extinction model can be expressed as:

$$\vec{N}_{t+1} = \vec{N}_t + G_t^+ \cdot \mathbf{K} - G_t^- \qquad (2)$$

At each step, G_t^-,the loss of abundance due to natural processes, is removed from the total abundance, and G_t^+ the abundance produced at time t, is redistributed across space by $\mathbf{K}$.

2.3 Accounting for sampling bias

Several limitations usually occurs in such colonization-extinction models, for example, carrying capacity abundances of the species across the study map are not known. Usually ecologists only have access to occurrences and estimated distribution extent. Hence $\vec{C}$ is not known at the beginning of the simulation.

The major problem with opportunistic occurrences data is the inconsistency of sampling effort across geographical scales (Johnston, Matechou, and Dennis 2023)(Bowler, Callaghan, et al. 2022) and taxonomy (Troudet et al. 2017). At the European scale for example, western Europe is tremendously more represented in citizen science networks -such as GBIF- compared to eastern Europe where occurrences are often scarce (Beck et al. 2014). This biases lead to difficulties to retrieve abundance from occurrences density. In this work we suppose the true abundance of a wider group of species shearing similar sampling pattern (hereafter called reference taxa) remains constant across the considered map. We further assume that the number of occurrences reported for a given species depends proportionally to the ratio between this species abundance compared to the total reference taxa abundance. We propose here to rely on the fact that a wider species pool shows less variation across space in comparison with a single species. This assumption is already done in other fields such as in environmental DNA, to compare relative abundances across samples. (see. example here (Sard et al. 2019)).

Using relative abundance defined by:

$$\tilde{N}_i(t) = \frac{N_i(t)}{C_{taxa}}$$

where C_{taxa} is an unknown constant quantity over the study area. We can rewrite the growth model only in term of relative abundance compared to C_{taxa}:

$$\Delta\tilde{N}_i(t) = g\frac{N_i(t)}{C_{taxa}}\left(\frac{\frac{C_i}{C_{taxa}} - \frac{N_i(t)}{C_{taxa}}}{\frac{C_i}{C_{taxa}}}\right) = g\tilde{N}_i(t)\left(\frac{\tilde{C}_i - \tilde{N}_i(t)}{\tilde{C}_i}\right) \qquad (3)$$

Leading to the same growth equation but using instead absolute abundance instead rel-

ative ones. One can see that $\tilde{C}_i = C_i/C_{taxa}$ -the relative carrying capacity for site i- is unknown and is a prerequisite to simulate the colonization-extinction dynamics in a relative manner. In our work, this quantity is assumed to be a function of the local habitat suitability h_i at this location, and is estimated using the following methodology:

We model the observation process as a binomial law, where the number of occurrences at a given location for a given time period, is directly proportional to its relative abundance $\tilde{N}_i$ - a.k.a the proportion of observable individuals of focal species compared to the entire number of observable individuals for the taxa- . Therefore, for a given number of occurrences $n_{taxa,i}$ for the entire taxa, only a smaller number of observations $n_{sp,i}$ corresponds to the focal species. Since the focal species is part of the entire taxa, $n_{sp,i} \leq n_{taxa,i}$, and $n_{sp,i} = n_{taxa,i}$ if the reference taxa only contains the focal species. The observation process can be viewed as:

$$X_i|\tilde{N}_i = \tilde{n}_i \sim Binomial(n_{taxa,i}, \tilde{n}_i)$$

where $\tilde{n}_i$ is the relative abundance of the focal species compared to the reference taxa and X_i the random variable giving the number of observations of the focal species. Since $X_i|\tilde{N}_i$ follows a binomial, when considering a uniform prior for $\tilde{N}_i$, it follows with respect to $X_i = n_{sp,i}$ its conjugate distribution: a Beta distribution with parameters:

$$\tilde{N}_i|X_i = n_{sp,i} \ \sim Beta(n_{sp,i} + 1, n_{taxa,i} - n_{sp,i} + 1)$$

2.4 Fitting carrying capacity

In this section, we call core range, a subset of the colonized range where abundance has reach an equilibrium. Based on the previous section we can estimate the posterior distribution of $\tilde{N}_i$ at each location i within the core current range distribution of the focal species. By definition (3), $\tilde{N}_i$ should be close to $\tilde{C}_i$ at equilibrium once colonization is complete. We used IUCN range extent as approximation of this core range, we assumed that the relative abundances reached an equilibrium that depends mainly on the local value of h_i. We aggregated all locations in the species' core range with at least one presence to build posterior distribution of relative abundance value for different values of h.

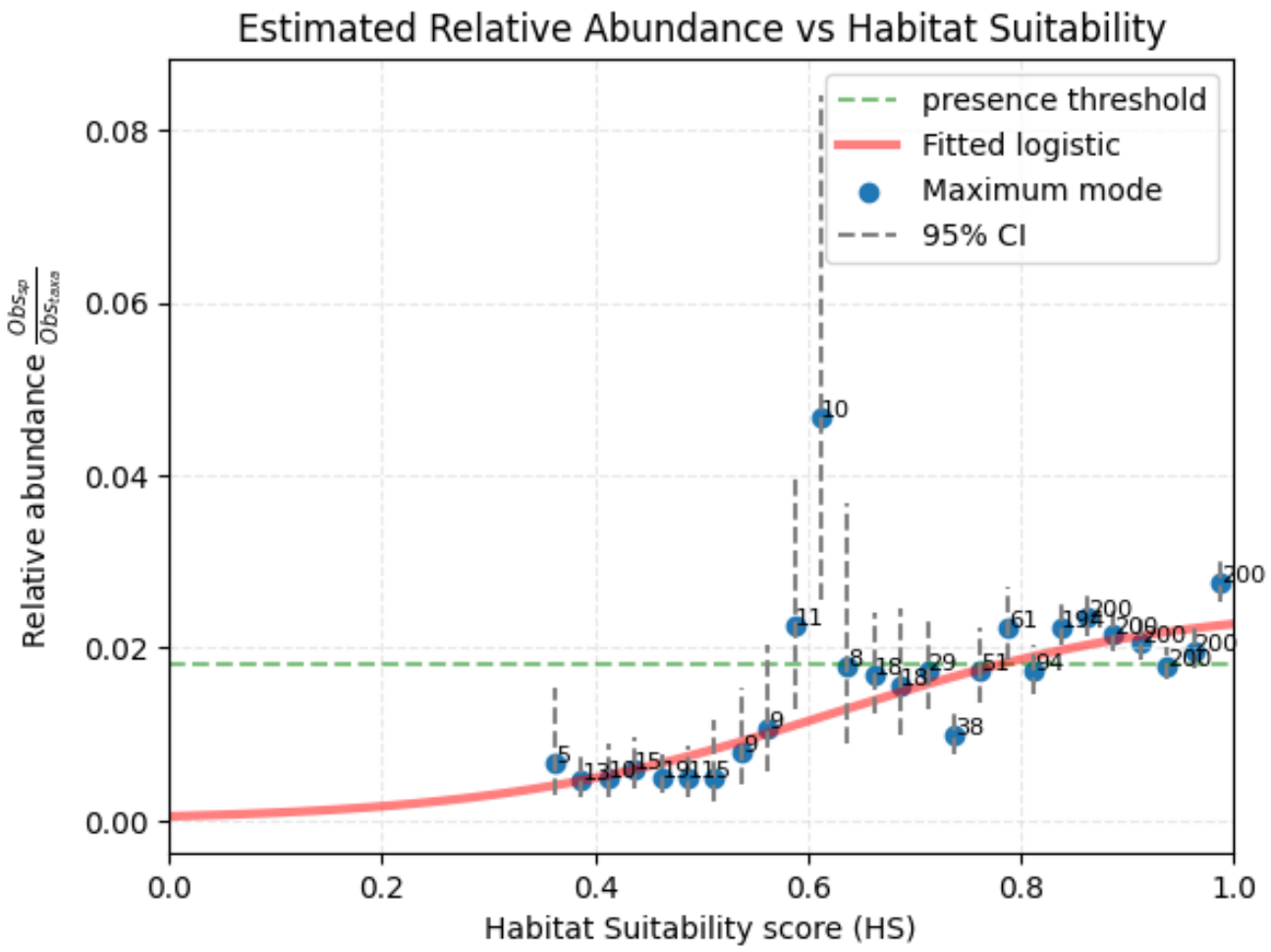


Figure 1: Estimated relative abundance compared to habitat suitability score h in core regions for the Black Winged Kite (*Elanus caeruleus*), numbers indicate the number of locations that has been used to estimate posterior distribution for each chunk.

We can see $\tilde{n}_i$ as the real unknown relative abundance, approximated by our observations at location i by its maximum likelihood estimator $n_{sp,i}/n_{taxa,i}$.

The relative simulation framework presents many advantages but rises a new problem: in conservation biology, protected areas cost computation and location optimization, rely on binary range extent of the modeled species. To deal with this limitation, the simulated abundances need to be converted into "presence" or "absence" and therefore a threshold of relative abundance corresponding to a "presence detection" must be defined. Since this definition is not absolute and depends on the modeling aims that favors sensitivity or selectivity (Manel, Williams, and Ormerod 2001), we propose to use the same definition of "presence" as the distribution model used to initialize the colonization-extinction simulation. We optimized a threshold by minimizing the mean squared error for presence/absence prediction from observed ratios $n_{sp,i}/n_{taxa,i}$ while using the initial distribution as ground-truth classification. The obtained threshold is the relative abundance above which the initial model distribution classify it as a presence (estimated as $\tilde{n} = 0.017$ for the black winged kite example see.fig. 1).

2.5 Reducing parameter set: coherence with mean natal dispersal distance

In our dispersal part, three parameters are necessary E_w, n, r, a last one, g need to be estimated for the growth part. To reduce the number of parameters, we investigate for relation between variables. E_w can be set respectively to r in order to ensure a mean dispersal distance MDD in a mean isotropic suitability landscape. The objective is to observe a mean dispersal distance of MDD across the range of the species, while allowing local discrepancies depending on the structure and nature of the local landscape.

In a uniform landscape without mortality, we expect the number of steps to displays a mean E_w. In a landscape with obstacles and not perfect habitat suitability we could expect mortality due to several processes. We assume these processes correlated with the habitat suitability score. Such that lower score means more mortality. In our model the survival probability is given by h^r. The fundamental transition is given by $\frac{E_w}{1+E_w}$, therefore the true corrected transition rate will be $\frac{E_w h^r}{1+E_w}$. Leading to a mean number of steps Ew^* lower than E_w due to the induced mortality.

$$\frac{E_w^*}{1+E_w^*} = \frac{E_w h^r}{1+E_w}$$

$$E_w = \frac{E_w^*}{h^r - E_w^*(1-h^r)} \qquad (4)$$

We can see that without mortality ($r = 0$) or perfect habitat suitability ($h = 1$) we retrieve $E_w^* = E_w$.

We now derive the explicit form of the distribution of establishing propagules starting from origin, in order to link their theoretic mean dispersal distance with MDD. The dispersion is modeled as a discrete random walk on a regular grid of $\mathbb{Z}^2$. The X and Y position of a particle starting in $(0,0)$ after L steps, can be viewed as a non-independent displacement shared between horizontal and vertical directions, probability $1/4$ for each up, down, right, left. But knowing Y and the number of steps L gives an information on X since displacement is either vertical or horizontal. As an example $Y = +L \implies X = 0$. On the orthonormal basis (e_X, e_Y) where e_X is the unitary horizontal vector and e_Y the unitary vertical vector, the final position is given by (X, Y):

$$L \sim Geom(p) \quad L_X \sim Binomial(\frac{1}{2}, L) \quad L_Y = L - L_X$$

$$X = \sum_{l=0}^{L_X} \pi_l \qquad Y = \sum_{l=0}^{L_Y} \pi_l \qquad \pi_l = \begin{cases} +1 & \text{with probability } \frac{1}{2}, \\ -1 & \text{with probability } \frac{1}{2}. \end{cases}$$

However, by changing the basis describing the position of the particle, we can create an independent displacement. We consider the basis (e_X^*, e_Y^*) such that, $e_X^* = \frac{\sqrt{2}}{2}(e_X + e_Y)$ and $e_Y^* = \frac{\sqrt{2}}{2}(-e_X + e_Y)$, where (e_X^*, e_Y^*) is basically a rescaled and rotated by $\pi/2$ radians version of (e_X, e_Y). In that basis, each displacement, vertical or horizontal always triggers the change along both axis of (e_X^*, e_Y^*) simultaneously. As an example: a single vertical displacement in the basis (e_X, e_Y) is described by $(X, Y) = (0, +1)$, in the basis (e_X^*, e_Y^*) it is described by $(X^*, Y^*) = (+1, +1)$. Respectively in the original basis down, right and left displacement are now described by $(-1, -1)$, $(+1, -1)$ and $(-1, +1)$. This change of basis permits to describe the entire random walk as two independent processes:

$$L \sim Geom(p) \qquad X^* = \sum_{l=0}^{L} \pi_l \qquad Y^* = \sum_{l=0}^{L} \pi_l \qquad \pi_l = \begin{cases} +1 & \text{with probability } \frac{1}{2}, \\ -1 & \text{with probability } \frac{1}{2}. \end{cases}$$

Where p is $\frac{E_w^*}{1+E_w^*}$, we can deduce the characteristic function of the final distribution of X^*, noted hereafter $\varphi_X^* = \mathbb{E}(e^{itX^*})$.

$$\varphi_X^*(t) = \mathbb{E}_L(\mathbb{E}(e^{itX^*}|L))$$

$$= \mathbb{E}_L(\mathbb{E}(e^{it\sum_l^L \pi_l})) = \mathbb{E}_L(\mathbb{E}(\prod_{l=0}^{L} e^{it\pi_l})) = \mathbb{E}_L(\prod_{l=0}^{L} \mathbb{E}(e^{it\pi_l})) = \mathbb{E}_L(\prod_{l=0}^{L} cos(t))$$

$$= \sum_{l=0}^{\infty} \mathbb{P}(L=l)cos^l(t) = \sum_{l=0}^{\infty} p^l(1-p)cos^l(t) = \frac{1-p}{1-pcos(t)}$$

Finally:

$$\varphi_X^*(t) = \frac{1-p}{1-pcos(t)}$$

and similarly for Y^*. We can recognize here the characteristic function of a discrete Laplace distribution where $\lambda = \frac{1-p}{\sqrt{1-p^2}}$. Hence, X^* follows a discrete Laplace distribution with parameter λ and similarly for Y^*. Then since the full distribution is given by a simple product cause X^* and Y^* are independent:

$$\mathbb{P}(X^* = x^*, Y^* = y^*) = \left(\frac{1-\lambda}{1+\lambda}\right)^2 \lambda^{|x^*|+|y^*|} \qquad (6)$$

Since the expected euclidean distance at establishment is intractable in discrete case, we approximate the discrete distribution product of 2 independent discrete Laplace, with its analogous continuous case: a product of two independent Laplace distributions where decays match our discrete case:

$$\mathbb{P}(X^* = x^*, Y^* = y^*) = \frac{1}{4b^2} e^{-\frac{1}{b}(|x^*|+|y^*|)}$$

where $b = -1/ln(\lambda)$.

In the continuous case and using radial coordinates, the expected distance is derived:

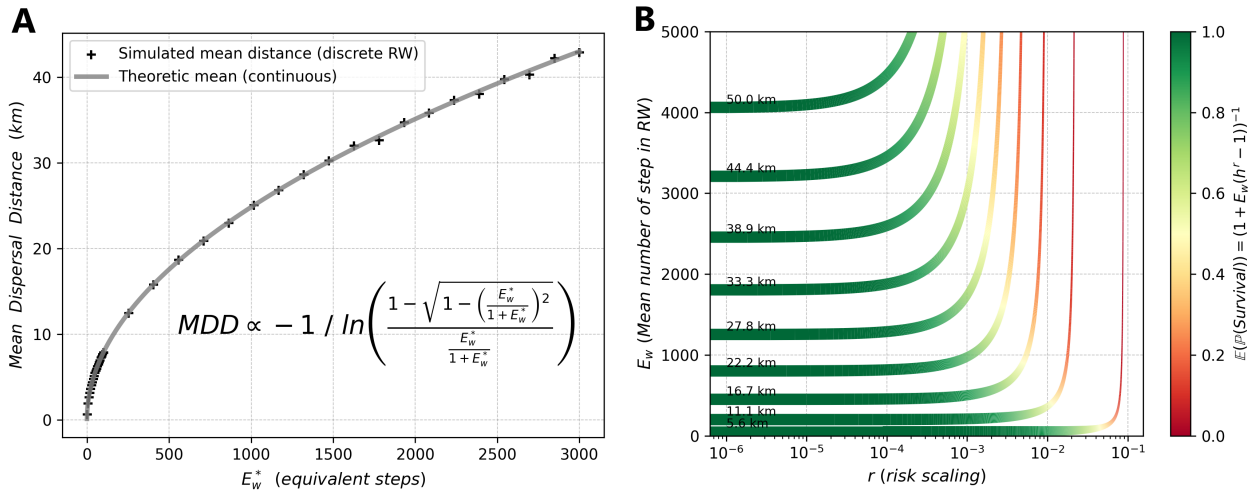


Figure 2: **A**) Link between the mean dispersal distance and the parameter E_w^* , showing the continuous approximation compared to discrete simulated data, α has been estimated as 1.11. **B**) Parameter E_w to ensure different mean dispersal distances MDD in an isotropic landscape with uniform habitat suitability $h = 0.8$ and risk scaling r. The color-mapping shows expected survival in such conditions.

$$\mathbb{E}\left(\sqrt{X^2+Y^2}\right) = \frac{\sqrt{2}}{2}\mathbb{E}\left(\sqrt{X^{*2}+Y^{*2}}\right) = \frac{\sqrt{2}}{2}\int_0^{\infty}\int_0^{2\pi}\frac{r^2}{4b^2}e^{-\frac{r}{b}(|cos(\theta)|+|sin(\theta)|)}drd\theta$$

By a change of variable : $r' = r/b$

$$= b \cdot \underbrace{\frac{\sqrt{2}}{2}\int_0^{\infty}\int_0^{2\pi} r'^2 e^{-r'(|cos(\theta)|+|sin(\theta)|)}dr'd\theta}_{\alpha} \quad \propto \quad b$$

The later result, permits to link directly E_w^* to the mean dispersal distance realized in an isotropic landscape with habitat suitability h and risk scaling factor r. This permits to ensure consistency between model parameters and the mean dispersal distance, ensuring that over the entire map, the mean dispersal distance would be close to MDD, despite being locally different depending on the surrounding landscape. In practice due to the continuous approximation the theoretic α is slightly higher than in the discrete case. We estimate directly α for the discrete case by linear regression (see. Fig.2A). We ensure MDD as the mean dispersal distance:

$$MDD = \mathbb{E}\left(\sqrt{X^2 + Y^2}\right) = \frac{-\alpha}{ln\left(\frac{1-\sqrt{1-\left(\frac{E_w^*}{1+E_w^*}\right)^2}}{\frac{E_w^*}{1+E_w^*}}\right)}$$

Therefore, knowing (4)

$$E_w = \left(h^r \frac{1 + e^{-2\alpha/MDD}}{2e^{-\alpha/MDD}} - 1\right)^{-1} \qquad (7)$$

We obtain here the relation between E_w and r parameters to ensure a known mean dispersal distance of MDD across the landscape. Permitting to reduce the parameter set to estimate to only three parameters: (n, r, g). Finally, we can see in Fig.2B, that beyond a certain r and for a given MDD no E_w permits to ensure a mean dispersal distance of MDD. This came from the fact that for a given MDD, higher r diminishes the mean dispersal distance reached, since more individuals are dying before establishment, therefore a higher E_w is needed to compensate the mortality. Beyond a certain point no higher dispersal capacity E_w permits to compensate the mortality. Hence, knowing a MDD value, gives directly a higher boundary for r to allow the model to be valid. This higher boundary is deduced from (7):

$$\left(h^r \frac{1 + e^{2\alpha/MDD}}{2e^{-\alpha/MDD}} - 1\right) > 0$$

$$r < \underbrace{\frac{1}{log(h)} log\left(\frac{2e^{-\alpha/MDD}}{1 + e^{-2\alpha/MDD}}\right)}_{r_{max}} \qquad (8)$$

(8) can therefore be used as a higher bound for the prior of r during learning. The value h is taken as the mean habitat suitability over the colonized range of the species, since we desire not to enforce MDD locally but across the whole landscape.

2.6 Joint estimation of dispersal and growth parameters

By expressing E_w as a function of MDD and r we reduced the parameter set to estimate to only three variables (n, r, g) and constraining our model to respect the known mean dispersal distance across the focal species core distribution. The 3-dimensional parameter set is then estimated using the assumed pseudo-equilibrium in the core-distribution of the species. We

choose a set of calibration sites where the objective was to explain the observed ratios distribution across these sites knowing that at equilibrium dispersal mortality and concentration effects should counteract growth effects. Leading to the equilibrium equality:

$$\vec{N}_\infty = \vec{N}_\infty + G_\infty^+ \cdot \mathbf{K} - G_\infty^- \qquad (4)$$

For these sites, starting at relative carrying capacities $\vec{\tilde{C}}$ we apply successively dispersal and growth steps to reach stationarity. G_∞ depends on g (see. Eq.2) and $\mathbf{K}$ depends on n and r (see. Eq.1). We measure the correspondence between simulated equilibrium for a parameter set (n, r, g) and the posterior distribution of observed ratios as:

$$\mathcal{L}(n, r, g) = -\sum_k \frac{1}{|\Omega_k|} \sum_{s \in \Omega_k} logL_s$$

where L_s is the likelihood of the simulated relative abundance compared to the one observed at location s. Ω_k is the set of selected location where a ratio $n_{sp,i}/n_{taxa,i}$ can be computed in a calibration site k. This cost function design uses for each sites its mean log-likelihood, that permits to rely on site complementarity rather than sampling discrepancies. In practice locations selection inside a calibration site ensures 50% prevalence between presence and absence locations. Absences are selected using a Disk-Poisson sampling method to reduce noise near presence locations. The calibration sites has been chosen with a size of 70×70 km, inside the core-distribution, with enough occurrences and distance between pairs of sites to avoid auto-correlation. This calibration size has been chosen large enough to allow for spatial abundance variation while be small enough to reduce computational speed. Sites are also selected based on their variability in term of habitat suitability values since flat h provides very restricted information on dispersal. Calibration sites are chosen outside simulated window, with at least 10 occurrences of the species. A variability of h sufficient to gain information on dispersal behavior is needed, to do so we conducted a binarization using Otsu thresholding (Otsu 1979) and selected sites with a inter / total variance ratio superior to 0.5, which allows for enough variability to extract dispersal information. The cost function $\mathcal{L}$ is then minimized using a mini-batch Adam optimizer (Kingma and Ba 2017) for approximately 2500 epoch, using at each new iteration a small batch of 3 random calibration sites among the dozen selected. The initial point in the parameter space is choose as the center of the priors for each parameters. Without further information on the focal species

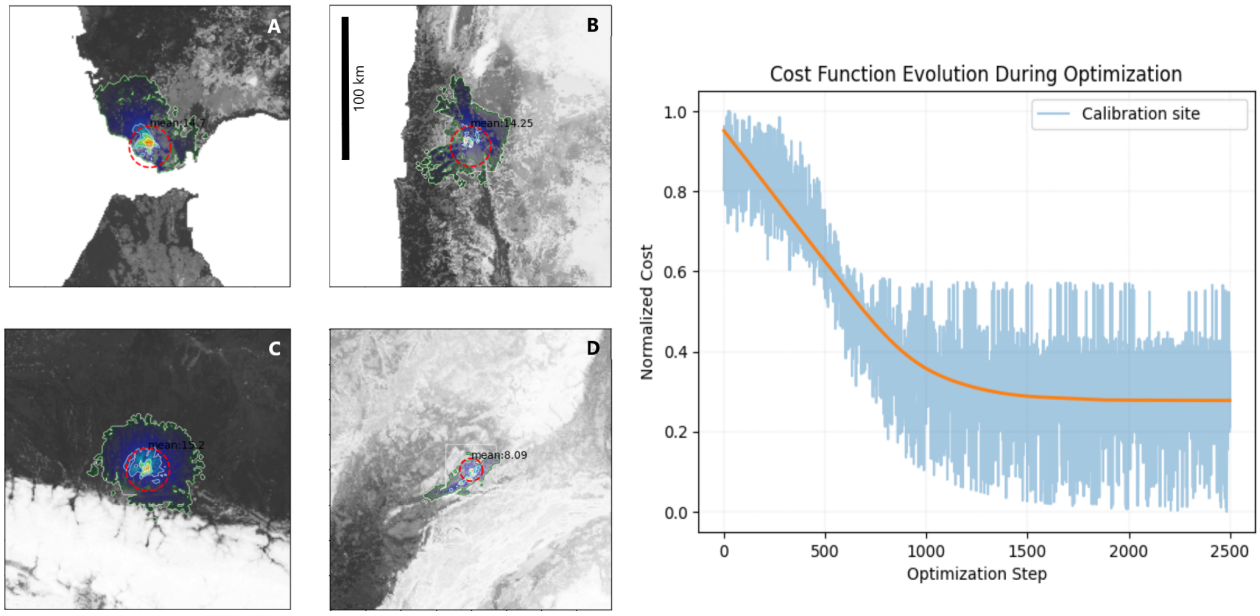


Figure 3: Left panels: Dispersal kernels with same dispersal parameters in different landscape contexts, underlying habitat suitability is shown in grayscale, kernel origin is located at the center of the window, green contour indicates area reachable by the species in one dispersal step. Red to blue colors indicate high to low probabilities to reach a given location from the origin. Red dotted circle indicate mean distance reached by propagules from the center. **Right panel:** Cost function $\mathcal{L}$ during optimization, the stochasticity is due to the mini-batch selection, here 3 calibration sites are chosen randomly among the dozen selected at each new step to compute $\mathcal{L}$. The optimization is run using Adam optimizer from PyTorch for 2500 epoch.

we advocate for flat priors $T_g \sim U[0, 10]$, $n \sim U[0, 3000]$ and $r \sim U[0, r_{max}(\bar{h})]$. With $\bar{h}$ is the mean habitat suitability score across occupied sites in the core-distribution, and T_g is the corresponding time steps needed to reach 95% of carrying capacity with growth rate g.

Fig.3, left panel shows example of dispersal kernels generated with same parameter set in different habitat suitability landscapes. Dispersal kernels Fig.3A and Fig.3B are clearly anisotropic, especially B that shows preferential directions in valleys of high habitat suitability. Kernel Fig.3C is less constraint by barriers allowing for a more circular shape. Kernel Fig.3 D is located in a radically difficult landscape impeding propagules displacement, the kernel is shrink towards the starting location, and the mean traveled distance is decreased.

3 Case study: The Black winged kite and European beaver, two European expanding species

3.1 The Black Winged Kite expansion in the South-West of France

The black winged kite (*Elanus caeruleus*), is a fast expanding species in the South-West of France (Mourguiart 2025). Arrived around the year 1990 from Spain (Cognet and Ballereau n.d.), the species quickly colonized the Western France littoral, and quickly expanded heading North. We conducted a reverse simulation, where we fitted the colonization-extinction model on calibration sites outside France but inside the known distribution of the species, considered as independent of the dynamics in France. Then, we predicted the expansion of the black winged kite in France from an approximated starting point in 1990 until the simulated year 2025 and compared the result to the current distribution. The mean natal dispersal distance MDD is evaluated as 21 $km/generation$ (Weeks et al. 2022).

We conducted 10 independent simulations to evaluate the variability of model outputs due to the selection of calibration sites. Fig.4 shows the expansion prediction over a long simulated window -more than 30 years- in coherence with the current observed distribution. PARADIS is able to retrieve the principal colonization front from the first occurrence point, despite the fact it only uses 1 input parameter -the mean dispersal distance- and readily available raster -occurrences and habitat suitability-.

More precisely current distribution is patchily distributed further from the main colonization front, likely due to long distance dispersal events, which constitute a long-time known limitation in dispersal modeling (Trakhtenbrot et al. 2005)(Nathan 2006)(Nathan et al. 2003) due to their rarity and stochasticity making them difficulty to predict. Due to these effects, the colonization-extinction model is predicted to under-estimate the true expansion, however this is less unlikely as the simulated time window is reduced, since such events have less chance to occurs during the simulated range. The slight under-prediction could also be due to our imprecise initial range that rely on scarce occurrences in the 1990, that are likely to under-estimate the true distribution at this time.

The main advance is that the model managed with minimal parametrization (only one

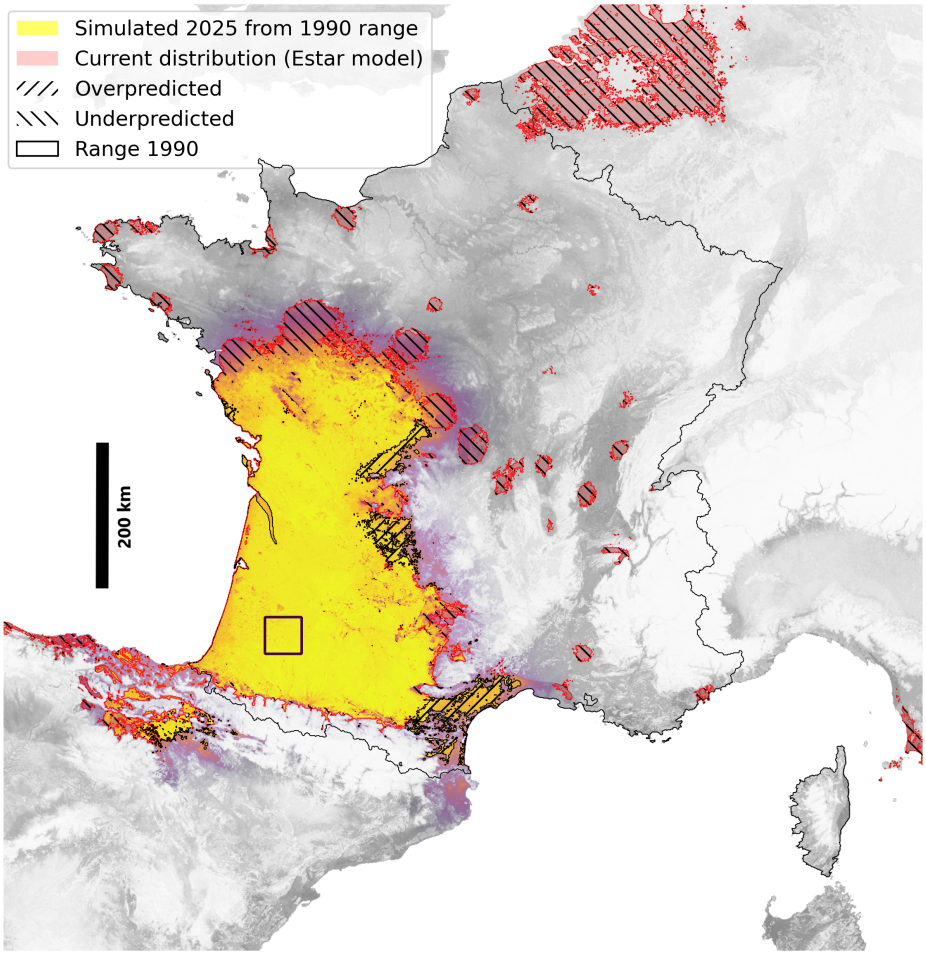


Figure 4: Simulated expansions from the black square region (1990 estimated first appearance) until simulated year 2025 are shown in yellow. Estimated current distribution with ESTAR framework [Hoareau et al. 2025] is shown in red, hashed regions corresponds to overpredicted or underpredicted regions by the colonization-extinction model. Dispersal parameters have been learned on independent calibration sites in Spain.

mandatory parameter) to reconstruct the main colonization shape, by predicting incapacity to cross the mountain chain of the Pyrenees in the southern front, and impossibility of long-term establishment in the mountainous cluster of Massif Central in central France. The model also managed to identify a correct expansion speed solely based on spatial occurrences data outside the simulated window.

3.2 European Beaver reintroduction in France

As a second example, we investigated the the European beaver (*Castor fiber*) case, an emblematic species of large fluvial areas. The European beaver nearly disappeared in Europe due to exploitation for its fur and castoreum (Halley and Rosell 2002). Its reintroduction began during the 1980's in France to expand its remaining population in the Rhone's river. Helped with dozens of reintroductions across decades and change of its protected status, the species has since recolonized major France's rivers (Dewas et al. 2012).

Once again, we conducted a similar reverse simulation as for the black winged kite, but accounting this time for breeding ground selection since the European beaver only breeds in large fluvial areas. This took form in our simulation as a spatial filter allowing reproduction only in major rivers up to three ramifications levels from bigger ones, while dispersal is still only constrained by suitability and dispersal capability as in the black winged kite example allowing individual exchanges through land between rivers.

In fig. 5 A we predicted using PARADIS fitted on spatial occurrences patterns outside France (Scandinavia, Spain and Eastern Europe) the expansion of the European beaver in France from its historical range in the Rhone river (1981) and knowing its reintroduction sites over the followings decades extracted from (Dewas et al. 2012). We show on the right panel fig. 5 B, the corresponding ground truth distribution observed in France in 2025. The model correctly predicted the overall colonization speed, especially in the Eastern part of its distribution, the model under-predicted its propagation along the Loire river, where the recolonization from the 1975 reintroduction point was faster than predicted both upstream and downstream. Finally the model correctly predicted the difficulty of the species to disperse from the 1969 reintroduction point in the north-west, likely caused by intensive agricultural landscape (Dewas et al. 2012). Overall PARADIS managed to predict a realistic range extension on an extensive simulated time range 1969-2025.

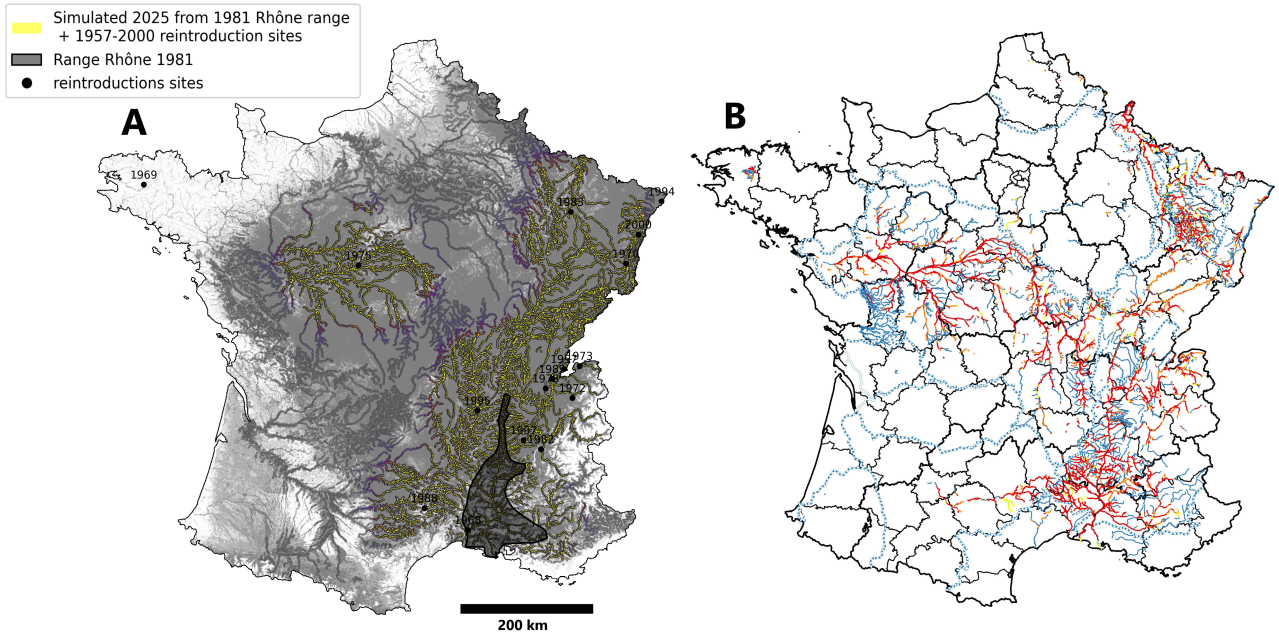


Figure 5: A) Simulated expansions from the Rhone's range (1981) and known reintroduction sites until simulated year 2025, habitat suitability is displayed in grayscale. B) Observed current distribution from castor observation network and French Biodiversity Office (blue: verified absence, yellow: possible presence, orange: likely presence, red: known presence, dotted blue : limits of water basins, dispersal parameters has been learned on occurrences pattern in Spain and Scandinavia, considered independent of dynamics in France.

4 Discussion

The model developed in this work addresses a central limitation identified in syntheses of species distribution and range-dynamics modelling: the lack of approaches that are simultaneously process-explicit, scalable, and parsimonious (Briscoe et al. 2019). PARADIS achieves this by relying on a deliberately low level of parametrization, requiring only readily available habitat suitability, opportunistic occurrences rasters and an estimate of mean natal dispersal distance. This contrasts with many existing dynamic range models, where realistic simulations of dispersal and population dynamics often depend on a large number of poorly constrained demographic and movement parameters.

Low parametrization has traditionally been associated with simple colonization–extinction models relying on isotropic dispersal kernels (Engler, Hordijk, and Guisan 2012) (Malchow, Bocedi, et al. 2021). Here, we demonstrate that a restricted and explicit set of assumptions allows dispersal processes to be reformulated in a way that preserves anisotropic, landscape-dependent propagation while maintaining a minimal parameter set. By embedding relative abundance dispersal and mortality during movement into a single anisotropic kernel, PAR-

Table 2: Comparison between different simulation methods

Model/software	Input layers	Total	User	Growth method	Dispersal method	Reference
RangeShiftR - expkern	HS	4	4	Logistic	Iso. kernel (exp)	Malchow, Bocedi, et al. 2021
poems	HS	4	4	Ricker	Aniso. kernel (LCK)	Fordham et al. 2021
MetaRange.jl	climatic layers	6	6	Beverton–Holt	Iso. kernel (exp)	Blechschmidt and Cabral 202
demoniche	HS	7	6	Age structured	Iso. kernel (exp)	Nenzén et al. 2012
CATS	HS	14	13	Age structured	Iso. kernel (exp)	Gattringer et al. 2023
PARADIS	**HS + op.occ**	**8**	**1**	**Logistic**	**Aniso. kernel (RWK)**	**this work**
Bayesian dynamic SDM	HS + op.occ (dated)	16^{+}	0^{+}	Age-structured	adjacent cells + LDD	Botella et al. 2022
steps	HS + barriers*	3	3	Age structured	Iso. kernel (exp)/ ind-based	Visintin et al. 2020
rangR	carrying capacity layer*	4	4	Ricker/Gompertz	Iso. kernel (half-t)	Markowska, Malinowska, and
MigClim	HS + barriers* + kern.*	6	6	Logistic/binary	Iso. kernel (exp)	Engler, Hordijk, and Guisan 2
RangeShiftR - CWR	HS + resistance layer*	6	6	Logistic	Ind-based CRW	Malchow, Bocedi, et al. 2021
RangeShiftR - SMS	HS + resistance layer*	7	7	Logistic	Ind-based SMS	Palmer et al. 2011
Mestre et al.	history of coloniz. extinct.*	5	0	Binary Markov	dist-based Hanski 1999	Mestre et al. 2017
SHIFT	HS + historical range*	2	1	Binary	Iso. kernel (inv-pwr)	Prasad et al. 2013
HSMM	abundance time series*	3	3	Ricker	Iso. kernel (d^2 exp)	Hooten et al. 2007
Bayesian × RangeShiftR	HS + abund. time series *	13^{+}	3^{+}	Age structured	Ind-based CRW	Malchow, Fandos, et al. 2024

Legend: Input: availability of necessary inputs prior simulation (see. legends) **Total:** Total numbers of parameters in the model except priors and hyper-priors.

User: number of parameters left to estimate by the user, typically account for mandatory parameters not automatically estimated in the method. **Growth method:** mathematical method used to simulate species natural processes (birth/death). **Dispersal method:** mathematical method used to account for species dispersal, when kernel-based method is used, the type of kernel is displayed. aniso. = anisotropic, iso. = isotropic, exp = exponential law, inv-power = inverse-power law, d^2 exp: exponential law using squared euclidean distance, RWK = Random Walk Kernel; CRW = Correlated Random Walk; SMS = Stochastic Movement Simulator; LCK = Least-Cost Kernel; LDD=long distance dispersal. The upperscript "$^{+}$" signals there is user-defined priors/hyperpriors additional to the parameters, the upperscript "*" denotes when a raster is hardly available.

ADIS avoids the need to explicitly specify quantities such as the number of propagules, stage-specific mortality rates, travel speed, or fine-scale behavioural rules. These parameters are frequently unavailable, difficult to estimate reliably, or require extensive expert knowledge and literature synthesis, especially for poorly studied species.

This design choice directly responds to concerns raised by Briscoe et al. 2019, who highlighted that many coupled SDM–population models are limited in practice by over-parameterization, identifiability issues, and strong reliance on expert-defined demographic inputs. PARADIS can be viewed as a parsimonious reformulation of the coupled SDM–population paradigm, in which spatial suitability constrains population dynamics while dispersal and growth processes are inferred jointly from observed spatial patterns rather than imposed a priori. As such, the model occupies an intermediate position between correlative SDMs, which scale well but lack explicit dynamics, and fully mechanistic or individual-based models, which offer high realism at the cost of tractability and generality.

It is important to note that some models aim is to learn directly from the occurrence record complex dynamic models thanks to a Bayesian framework (Botella et al. 2022), and left few parameters for the user to estimate which is a similar approach to our work. However this method has several limitations, in case of poorly sample species, the historical record can be too restricted to learn expansion dynamic from them. In contrast PARADIS learns not on the history of species expansions but on a static image of the occurrence record, allowing to aggregate several years to strengthen learning. Similarly, the high number of parameters in such model reduce their identifiability, exemplified by leveraged concerns on sampling chain convergence and high uncertainty on posterior distributions (Botella et al. 2022). The transferability of such model relies on the realism of their priors and hyper-priors, which are ultimately estimated through literature or experts, reducing its scalability. In contrary, despite being able to simulate remarkably diverse types of expansion dynamics, our model relies on minimal number of parameters, which seems enough to simulate accurately large scale expansion patterns while reducing the pressure on finding realistic priors, allowing us to use flat priors on theoretical ranges (see. Eq 8). Furthermore, our case studies show great potential for transferability since parameters were not been learn on the simulated window, despite being able to reconstruct realistically range expansions.

A further strength of the approach is its ability to simulate relative abundances consis-

tently linked to presence–absence definitions through a simple observation model. Calibration explicitly accounts for the presence definition of a reference distribution model, allowing the user to tune the balance between specificity and sensitivity according to the intended application (Manel, Williams, and Ormerod 2001). This ensures that binarized predictions remain coherent with both current observations and future projections, a point often overlooked in dynamic range modeling frameworks.

Importantly, PARADIS leverages widely available opportunistic occurrence data (e.g. GBIF, iNaturalist) to calibrate dispersal processes. While such datasets are known to suffer from strong spatial and taxonomic sampling biases [Troudet et al. 2017][Bowler,Callaghan 2022], the relative modelling framework used here reduces these biases by explicitly accounting for local sampling effort. This is made possible by comparing focal species records to those of a broader, ecologically comparable taxonomic group, allowing relative abundances to be inferred across sites. Similar assumptions underpin relative comparisons in environmental genetics and community-level analyses, and provide a pragmatic solution in data-limited contexts, despite long-time known limitations [Fonseca 2018].

Comparing to other simulation methods with similar level of realism: PARADIS uses significantly less species specific parameters or hardly available rasters (see. Table2). Overall, PARADIS directly addresses several methodological gaps highlighted by Briscoe et al. 2019: it incorporates explicit dispersal and colonization–extinction dynamics, remains applicable at large spatial scales, and minimizes data requirements. While the model does not aim to replace highly detailed demographic or individual-based approaches, it offers a tractable alternative for large-scale biodiversity modelling where such data are unavailable, thereby expanding the range of species and systems that can be studied dynamically.

4.1 Limitations and scope of application

PARADIS growth part is kept simple as possible, however, population dynamics espacially genetic and demographic effects at low densities can have an important role in species expansion [Phillips et al. 2010]. PARADIS is not taking into account Alee effect [Stephens et al. 1999] nor evolution of dispersal due to spatial genetic sampling [Phillips et al. 2010] or counter-selection by patch fragmentation [Cote et al. 2017], that were been shown to affect extinction patterns and colonization speed. Such effects can be modeled at the cost of

additional parametrization by individual-based models such the ones used by RangeshiftR (Malchow, Bocedi, et al. 2021).

Moreover, rare long-distance dispersal events are not explicitly part of PARADIS. Although infrequent, such events can disproportionately influence range expansion, particularly in invasive species where human-mediated transport along road networks or trade routes can dramatically accelerate spread and cannot be neglected [Hulme 2009]. Vector-mediated dispersal has been shown to increase invasion front velocities by several orders of magnitude and can generate highly discontinuous colonization patterns, especially under multiple introduction scenarios (Valentin et al. 2017). Such human-mediated dispersal, long-distance dispersal or jump-dispersal can be accounted, again at the cost of supplementary parameters, this is the case in (Botella et al. 2022).

By construction, PARADIS focuses on short-range dispersal driven by the intrinsic movement capacity of individuals and is therefore most appropriate for species or time horizons where long-distance jump dispersal can reasonably be neglected. In this sense, the model provides a conservative estimate of future range expansion, corresponding to a lower bound of expected spread. The true realized range, when accounting for stochastic jump events or anthropogenic vectors, is likely to be larger.

Rather than a downside, this property allows PARADIS predictions to be interpreted as a mechanistically grounded baseline scenario, complementary to the commonly used "no dispersal" and "full dispersal" assumptions (Bateman et al. 2013). As such, PARADIS can serve as a default realistic lower-bound scenario for species expansion, facilitating comparison across modelling approaches and improving transparency in forecasting exercises.

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